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Risk-predictive trajectory and speed planning method for safe autonomous driving on urban narrow roads

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Abstract
This study proposes a risk-predictive path planning and control framework for autonomous driving (AD) and advanced driver-assistance systems (ADAS) operating on narrow urban roads with limited visibility. When driving through unsignalized intersections, buildings or parked vehicles often create blind spots where pedestrians or other road users may suddenly dart out, becoming a significant safety risk. To address this challenge, this research defines a method to simultaneously determine the safe speed and safe lateral distance based on the geometric relationship between the ego vehicle and the occluded area. The safe speed is defined as the maximum velocity at which a collision can be avoided through emergency braking, while the lateral gap ensures avoidance of side-faced collisions. Using these risk-predictive parameters, a non-linear polynomial path generation method is formulated, in which cubic and quartic functions describe the vehicle's speed profile and trajectory, respectively. The coefficients of these polynomials are optimized to minimize deviations from desired paths while preventing the vehicle from entering the predicted high-risk zones by the proposed method. Comprehensive MATLAB simulations were performed using a 3-DoF vehicle model to validate the proposed method. Results show that the framework successfully generated smooth and safe trajectories, allowing the vehicle to avoid potential collisions under various road widths and darting-out conditions. Even in constrained road scenarios, the system maintained stability and avoided frontal collisions. The proposed method effectively balances safety and mobility, demonstrating strong adaptability for future integration into real-world AD/ADAS systems.

Keywords : Autonomous driving, Intersection safety, Potential risk, Active safety, Path planning, Narrow roads

1. Introduction

In recent years, extensive research has been conducted on path planning and speed profiles generation to realize Autonomous Driving (AD) and Advanced Driver Assistance Systems (ADAS) in urban environments (González et al., 2016; Paden et al., 2016; Reda et al., 2024). Driving on narrow roads in mixed traffic with vulnerable road users (VRUs) presents particular challenges for autonomous driving vehicles (Wang et al., 2022; Koç et al., 2021). When drivers navigate through unsignalized intersections on narrow roads, there is a risk of unexpected darting out of other road users, such as pedestrians, bicycles, motorbikes, or cars. This risk is further highlighted when crossing roads are occluded by objects such as buildings and parked vehicles (Wang et al., 2023). To avoid or mitigate collisions with objects that suddenly emerge from blind areas, human drivers and the AD/ADAS systems must decelerate to a safe speed through appropriate risk prediction (Wang et al., 2022; Brycht, 2020; Lin et al., 2023). For AD/ADAS vehicles to safely pass through intersections, they must decelerate in advance, but as slowly as possible. However, users often find such cautious behavior unacceptable due to perceived loss of comfort or mobility (Wang et al., 2023; Ito et al., 2022).

Observations of careful and competent human drivers show that they maintain sufficient lateral distance from occlusions, allowing them to reduce excessive deceleration. Our previous work defined safe speeds for intersection right-turns to balance safety and mobility, based on geometric relationships between the ego vehicle and occluding objects. However, this earlier research focused solely on intersection right-turn scenarios and predicted smooth, feasible

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trajectories (Fujinami and Raksincharoensak, 2021; Fujinami et al., 2021; Fujinami, 2021).

This study expands the scope beyond intersection turning to include driving through intersections, negotiating curves, interacting with surrounding dynamic objects, and overtaking parked vehicles. Our research is to define safe speeds and appropriate lateral gaps in risk-predictive scenarios that ensure both safety and high mobility in the residential area involving occluded dynamic VRUs and vehicles.

Focusing on straight driving on urban narrow roads, this paper proposes a risk prediction framework that simultaneously determines safe speed and lateral gap to avoid potential darting-out objects from blind spots. Furthermore, this paper introduces a path generation method that enables the vehicle to navigate smoothly without entering predicted high-risk areas by solving an optimization problem. The effectiveness of our proposed method is demonstrated through comprehensive simulation studies.

2. Risk definitions

This study aims to reduce collisions with the dynamic objects that may dart out from behind stationary obstacles in scenarios such as intersections with poor visibility, overtaking parked vehicles, and similar urban situations. Unlike conventional risk definitions based on probability and damage severity, the proposed framework predicts potential darting-out risk in a deterministic manner by identifying reachable collision regions of virtual darting-out objects under conservative assumptions. For clarity and simplicity, we first introduce a representative scenario in which the ego vehicle drives straight through an unsignalized intersection with poor visibility. Although this paper focuses on this specific case, the proposed method can be extended to other scenarios by appropriately setting the parameters that define risk. The geometry of the target intersection is shown in Fig. 1. The intersection is a T-junction where a narrow side road merges perpendicularly from the left onto the main road. A potential object may dart out from the side road and cross the main lane.

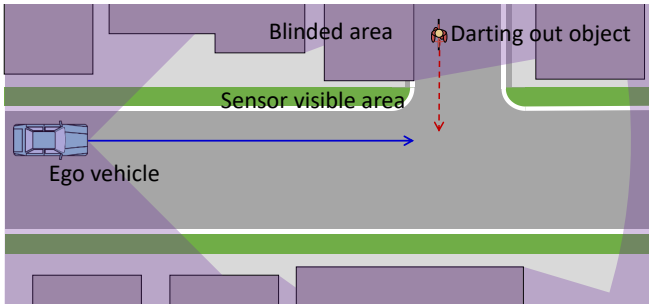


Fig. 1 Target unsignalized T-junction with poor visibility.

A key issue when a vehicle passes through an unsignalized intersection with poor visibility is whether the driver or system can avoid a collision by deceleration if an object suddenly darts out from the blind area. Drivers typically decelerate to an appropriate speed to anticipate such sudden crossing. However, if a darting out object strikes the side-face of the ego vehicle, the collision cannot be avoided merely by braking, even when equipped with an Autonomous Emergency Braking System (AEBS). On the other hand, maintaining a sufficient lateral distance from the occluded area can prevent side-face collisions. Accordingly, this section defines two primary risk factors for blind-spot scenarios: safe speed and safe lateral gap. The following subsections introduce these risk factors and discuss their characteristics.

2.1 Safe speed definitions

Previous studies defined safe speed as the speed at which an ego vehicle can stop via emergency braking before reaching a predicted collision point. However, this definition assumes a full stop and does not allow the ego vehicle to cross the potential trajectory of a darting-out object. To address this issue, earlier research introduced a virtual darting-out object and used a time-to-collision (TTC) threshold to evaluate visible distance at blind spots. When the visible distance exceeds a critical TTC (t_{crit}), the system does not intervene, because the darting-out object would also have sufficient time to avoid a collision. Therefore, safe speed guarantees collision avoidance particularly when visibility at blind spots is limited (Fujinami and Raksincharoensak, 2021; Fujinami et al., 2021; Fujinami, 2021). In this paper, a bicycle is considered as a representative virtual darting-out object, and t_{crit} = 1.5 s is adopted as a sufficient time margin for collision avoidance. This value represents the time required for a cyclist to perceive an approaching vehicle and

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complete an avoidance or stopping maneuver under typical urban conditions.

Building upon these definitions and considering a typical Japanese blind intersection as an example, we assume a virtual object darting out from an occluded area, as illustrated in Fig. 2. The ego vehicle is assured to be equipped with an object detection sensor located at the center of the front face of the vehicle. This research extends the concept to straight-driving scenarios at perpendicular intersections. Given the visibility at blind spot and the assumed darting-out trajectory, we can express the dart-out distance D_{crit} at TTC of virtual object t_{crit} and assumed darting-out speed of the virtual object v_{vir} as indicated in Fig. 2.

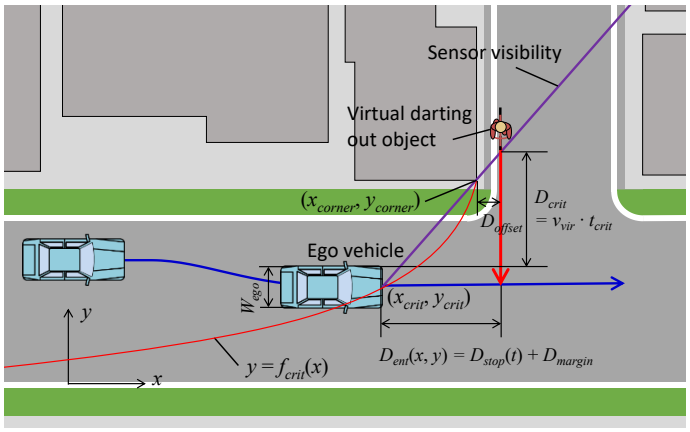


Fig. 2 Schematic definitions of the distances for safe speed calculation.

From these definitions, the position of the ego vehicle where the lateral darting-out distance of the virtual object from the blind spot becomes a certain distance D_{crit} can be expressed as a curve. The curve $y = f_{crit}(x)$ for the position of the ego vehicle (x_{crit}, y_{crit}) where $D_{crit} = v_{vir} \cdot t_{crit}$ as described in Eq. (1), since the position of the ego vehicle is geometrically related to the visible angle.

$$y = f_{crit}(x) = -\left(D_{crit} + \frac{W_{ego}}{2}\right)\frac{x - x_{corner}}{x - x_{corner} - D_{offset}} + y_{corner} \tag{1}$$

where (x_{corner}, y_{corner}) is the position of the corner, W_{ego} is the width of the ego vehicle, and D_{offset} is the offset distance from the corner to assumed virtual trajectory.

By defining the distance from position (x_{crit}, y_{crit}) to the assumed darting-out trajectory as $D_{ent}(x_{crit}, y_{crit})$, we can use the following equation to determine the safe speed for avoiding virtual darting-out objects:

$$v_{safe}(x, y) = \tau a_{EB} + \sqrt{-2a_{EB}\left(D_{ent}(x, y) - D_{margin}\right) + \tau^2 a_{EB}^2} \tag{2}$$

where a_{EB} is the acceleration of AEBS, τ is the system latency, and D_{margin} is the safety margin for AEBS.

As a characteristic of the curve, a larger lateral gap from the corner corresponds to a higher safe speed. Decelerating the vehicle speed to the safe speed along the curve $y = f_{crit}(x)$ ensures collision avoidance with respect to frontal collisions caused by darting-out objects whose speed is lower than v_{vir}.

However, if the safe speed were strictly enforced until the ego vehicle reaches the predicted dart-out trajectory, the safe speed would eventually decrease to zero as the vehicle approaches the blind spot, making it impossible to pass through the intersection. Therefore, a termination condition for the safe-speed-based control is required.

In this study, once the visible distance exceeds D_{crit}, that is, when the ego vehicle reaches the curve $y = f_{crit}(x)$, the occluded area is considered to be sufficiently observable, and there is no risk in terms of frontal collision risk. Beyond this curve, a darting-out object moving at a speed lower than v_{vir} cannot collide with the front face of the ego vehicle by definition.

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Nevertheless, even after the ego vehicle passes the curve $y = f_{crit}(x)$, a residual risk of side-face collision may still exist if a darting-out object emerges and moves with constant velocity. In such cases, although increasing t_{crit} provides the darting-out object with sufficient time for evasive action, it is difficult for the ego vehicle itself to actively avoid side-face collisions solely by braking. Therefore, the next subsection introduces an additional geometric condition to determine the optimal position (x_{crit}, y_{crit}) that enables the ego vehicle to avoid both frontal and side-face collisions.

2.2 Safe lateral gap definitions

By maintaining a sufficient lateral distance from blind spots, the ego vehicle can guarantee avoidance of side-face collisions and safely traverse occluded scenarios without collision risk. To determine the lateral gap for the ego vehicle passing through the intersection, we defined the possible darting-out distance from blind areas. Figure 3 illustrates the possible darting-out area of virtual objects. The green trapezoidal region represents the area that a virtual object can reach at vehicle speed $v_{ego}(t)$. Avoiding this trapezoidal area ensures prevention of side-face collisions with predefined virtual darting-out objects.

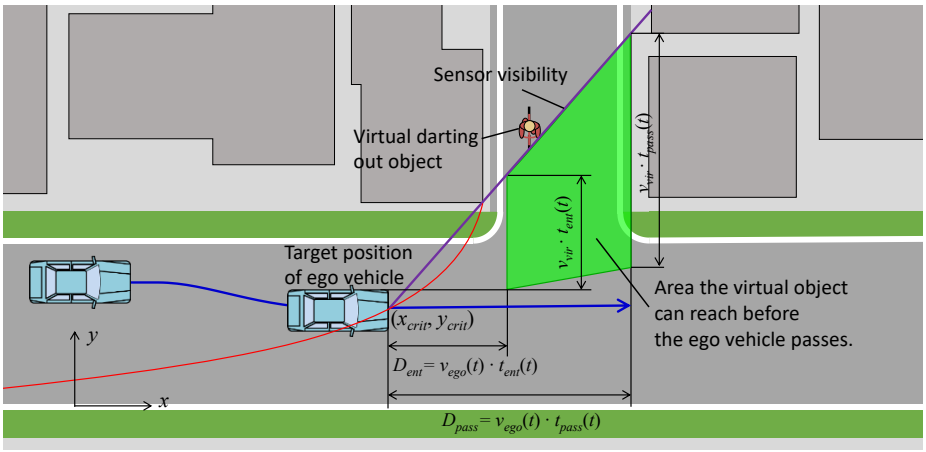


Fig. 3 Schematic definitions of darting-out area for virtual object.

Figure 4 shows the relationship between darting-out areas and safe speeds with respect to vehicle speed $v_{ego}(t)$. In this figure, the red line indicates curve $y = f_{crit}(x)$; however, the safe speed on this curve varies with positions. Therefore, we can represent a safe-speed curve for position (x_{crit}, y_{crit}) in three-dimensional space. Vehicles must travel at speeds lower than the defined safe-speeds surface, meaning the vehicles cannot pass through the red shaded regions.

To ensure both safety and mobility, the system should select the highest feasible speed with minimal lateral deviation. As shown in Fig. 4, we can simultaneously define the optimal safe speed v_{safe} and lateral position y_{crit}. If other road users or the limited road width restricts available lateral gaps and the ego vehicle needs to enter the green area, our method selects the maximum possible gap. Even in such cases, the vehicle can still avoid the frontal collision with the darting-out objects.

3. Non-linear polynomial path planning

The most significant advantage of this method is its high adaptability, allowing for the easy and simultaneous determination of both safe speed and safe lateral distance. By using the derived position (x_{crit}, y_{crit}) and the safe speed v_{safe} as waypoints for path planning, the vehicle path is generated by optimizing the coefficients of cubic and quartic polynomial functions, which determine the ego vehicle's speed profile and trajectory curve, respectively. Huy Do (2015). adopted a similar approach, utilizing curves defined by polynomial functions. That research expressed these functions as time-dependent, whereas the present study defines them as functions of position to facilitate waypoint adaptation. The objective of these functions is to ensure that the resulting curves do not enter restricted regions defined by factors such as speed limits, safe speeds, road boundaries, and potential darting-out areas. Figure 5 shows example curves of the speed profile and trajectory. The longitudinal position, speed, and lateral position at the terminal point of the curves are denoted as L_p, v_{Lp}, and y_{Lp} respectively. These paths are repeatedly generated at each control sampling time to compute the vehicle control commands.

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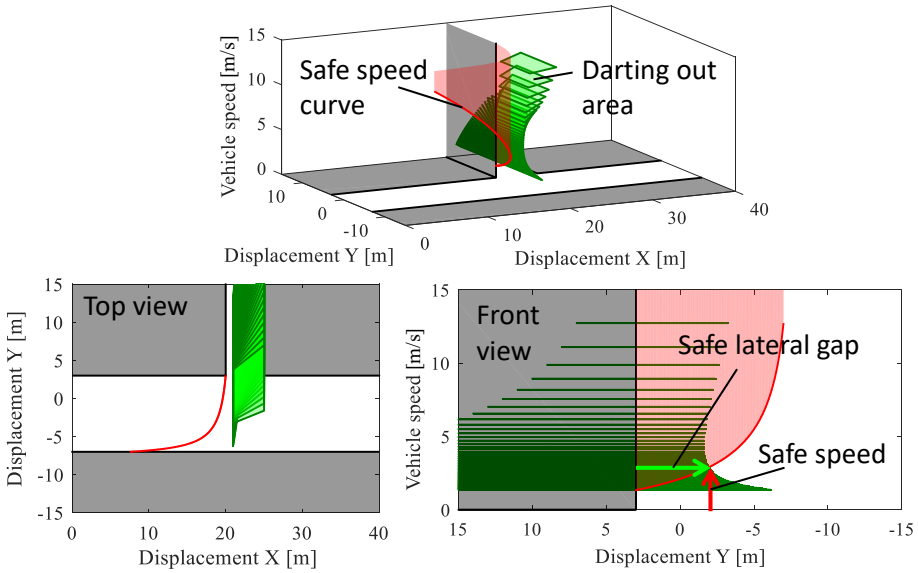


Fig. 4 The safe speed and darting-out area at the position and speed of the vehicle, and the definitions of safe speed and safe lateral gap.

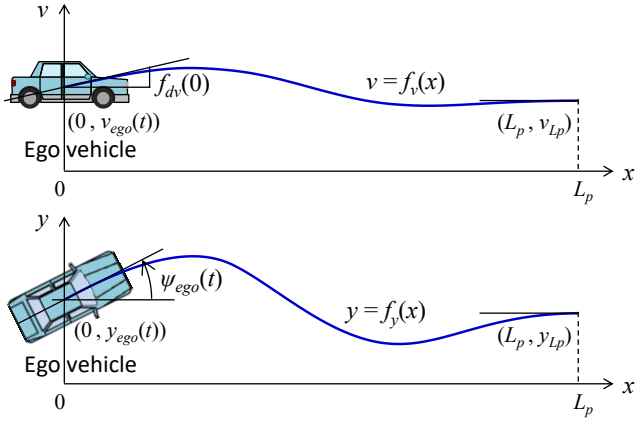


Fig. 5 Definitions of variables used to generate the curves. The positions and orientation of the x-axis origin represent the current position of the ego vehicle and the direction of the road, respectively. The look-ahead distance of the path, L_p, is defined as $L_p = v_{ego}(t) \cdot t_p$, where t_p is a constant look-ahead time.

3.1 Trajectory curve generation

The trajectory curve aims to generate a continuous and smooth curve which does not enter the unsafe area without crossing the road boundaries and collision predicted area. The trajectory is generated as fourth-order polynomial curve as Eq. (3). The fourth-order curve represents smooth steering motion, enabling the vehicle to bypass collision-predicted regions and subsequently return to the desired trajectory.

$$y = f_y(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0 \tag{3}$$

where, a_1 to a_4 are coefficients of the curve.

The first and second derivatives of Eq. (3) with respect to x are given by:

$$\frac{dy}{dx} = f_{dy}(x) = 4a_4 x^3 + 3a_3 x^2 + 2a_2 x + a_1 = \tan(\psi(x)) \tag{4}$$

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$$\frac{d^2y}{dx^2} = f_{ddy}(x) = 12a_4x^2 + 6a_3x + 2a_2 \tag{5}$$

where the yaw angle of the ego vehicle is ψ.

The yaw rate $\dot{\psi}$ of ego vehicle is:

$$\dot{\psi} = \frac{dx}{dt} \cdot \frac{1}{1 + (dy/dx)^2} \cdot \frac{d^2y}{dx^2} \tag{6}$$

From Eqs. (4)–(6), the coefficient a_2 can be derived from Eq. (7), considering the measured speed and yaw rate of the ego vehicle.

$$\begin{aligned} &\dot{\psi}_{ego} = \frac{dx}{dt} \cdot \frac{1}{1 + \left(f_{dy}(0)\right)^2} \cdot f_{ddy}(0) = v_{ego} \frac{2a_2}{1 + a_1^2} \\ &\Leftrightarrow a_2 = \frac{1 + a_1^2}{2} \cdot \frac{\dot{\psi}_{ego}}{v_{ego}} \end{aligned} \tag{7}$$

The boundary conditions at the terminal position of the curve are expressed as:

$$\begin{cases} f_y(L_p) = a_4L_p^4 + a_3L_p^3 + a_2L_p^2 + a_1L_p + a_0 = y_{L_p} \\ f_{dy}(L_p) = 4a_4L_p^3 + 3a_3L_p^2 + 2a_2L_p + a_1 = 0 \end{cases} \tag{8}$$

From the boundary conditions in Eqs. (3)–(5), and (8), the coefficients a_0 to a_4 can be derived as shown in Eq. (9).

$$\begin{cases} a_0 = f_y(0) = y_{ego} \\ a_1 = f_{dy}(0) = \tan(\psi_{ego}) \\ a_2 = \dfrac{f_{ddy}(0)}{2} = \left\{1 + \tan^2(\psi_{ego})\right\} \dfrac{\dot{\psi}_{ego}}{2v_{ego}} \\ a_3(y_{Lp}) = \dfrac{1}{L_p^3}\left(4y_{Lp} - 2a_2L_p^2 - 3a_1L_p - 4a_0\right) \\ a_4(y_{Lp}) = -\dfrac{1}{L_p^4}\left(3y_{Lp} - a_2L_p^2 - 2a_1L_p - 3a_0\right) \end{cases} \tag{9}$$

where y_{ego} is the lateral position of the ego vehicle. The coefficients a_3 and a_4 are determined by fixing the lateral position at the terminal point of the curve, y_{Lp}.

Thus, the trajectory curve function at longitudinal position x and lateral position y_{Lp} at terminal can be described as:

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$$f_y\left(x, y_{L_p}\right) = a_4\left(y_{L_p}\right)x^4 + a_3\left(y_{L_p}\right)x^3 + a_2x^2 + a_1x + a_0 \tag{10}$$

3.2 Speed planning

The speed profile curve aims to generate a continuous and smooth velocity profile that does not exceed the speed limit. The trajectory is represented by a third-order polynomial curve, as shown in Eq. (11). This cubic curve describes the smooth acceleration and deceleration necessary for the vehicle to reach both the safe speed and the desired cruising speed.

$$v = f_v\left(x\right) = b_3x^3 + b_2x^2 + b_1x + b_0 \tag{11}$$

The first and second derivatives of Eq. (11) with respect to x are as follows:

$$\frac{dv}{dx} = f_{dv}\left(x\right) = 3b_3x^2 + 2b_2x + b_1 \tag{12}$$

$$\frac{d^2v}{dx^2} = f_{ddv}\left(x\right) = 6b_3x + 2b_2 \tag{13}$$

Using Eqs. (11) and (13), the boundary conditions at the terminal position of the curve are expressed as:

$$\begin{cases} f_v\left(L_p\right) = b_3L_p^3 + b_2L_p^2 + b_1L_p + b_0 = v_{Lp} \\ f_{ddv}\left(L_p\right) = 6b_3L_p + 2b_2 = 0 \end{cases} \tag{14}$$

From the boundary conditions in Eqs. (11), (12), and (14), the coefficients b_0 to b_3 are derived as shown in Eq. (15).

$$\begin{cases} b_0 = f_v\left(0\right) = v_{ego} \\ b_1 = f_{dv}\left(0\right) = \dfrac{\dot{v}_{ego}}{v_{ego}} \\ b_2\left(v_{Lp}\right) = \dfrac{3}{2L_p^2}\left(v_{Lp} - b_1L_p - b_0\right) \\ b_3\left(v_{Lp}\right) = -\dfrac{1}{2L_p^3}\left(v_{Lp} - b_1L_p - b_0\right) \end{cases} \tag{15}$$

where the coefficients b_2 and b_3 are determined by fixing the speed at the terminal point of the curve, v_{Lp}.

Thus, the speed profile curve at longitudinal position x and terminal speed v_{Lp} at terminal can be described as:

$$f_v\left(x, v_{L_p}\right) = b_3\left(v_{L_p}\right)x^3 + b_2\left(v_{L_p}\right)x^2 + b_1x + b_0 \tag{16}$$

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3.3 Profile generation by solving optimization problems

The objective of the optimization described in this section is to generate a feasible trajectory and speed profile that satisfy predefined safety constraints while maintaining smooth and natural vehicle motion. The target trajectory and speed profiles are determined by solving an optimization problem. The cost function is defined in Eq. (17), and the boundary conditions of the functions are defined in Eqs. (18)−(22). The cost function evaluates the errors between the generated profiles and their reference profiles, as well as deviations from acceleration, speed, and drivable space limitations. The speed reference $v_{ref}(x)$ is defined from the legal speed limits. Between the calculated position x_{crit} and the end of the intersection, the speed reference $v_{ref}(x)$ is defined as the safe speed v_{safe}. Similarly, the reference line $y_{ref}(x)$ is defined as the road reference line, and between the calculated position x_{crit} and the end of the intersection, the reference line $y_{ref}(x)$ was defined as $y_{crit}(x)$. Therefore, in the proposed framework, safety requirements, such as collision avoidance, road boundaries, speed limits, and physical acceleration constraints, are imposed as hard constraints. These constraints strictly prohibit the vehicle from entering predicted collision-risk areas or violating physical or regulatory limits.

The variables y_{Lp} and v_{Lp} are obtained by solving cost function $z(v_{Lp}, y_{Lp})$ using the Levenberg-Marquart algorithm (Levenberg, 1944; Marquardt, 1963) to minimize the cost function $z(v_{Lp}, y_{Lp})$. By solving for y_{Lp} and v_{Lp}, the parameters in Eqs. (10) and (16) can be determined.

$$\begin{aligned} z\left(v_{L_p}, y_{L_p}\right) = \sum_{k=1}^{n} \Big\{ & w_1 \left| f_v\left(x(k), v_{L_p}\right) - v_{ref}\left(x(k)\right) \right| + w_2 g_v\left(x(k), v_{L_p}\right) + w_3 g_a\left(x(k), v_{L_p}\right) \\ & + w_4 \left(\left| f_y\left(x(k), y_{Lp}\right) - y_{ref}\left(x(k)\right) \right| \right) + w_5 g_y\left(x(k), y_{Lp}\right) + w_6 g_{ay}\left(x(k), y_{Lp}, v_{Lp}\right) \end{aligned} \tag{17}$$

where from w_1 to w_6 are weighting coefficients of the cost function. Each term of the equation represents, respectively, the error between the speed profile and the reference speed, the excess relative to the reference speed, the deviation from longitudinal acceleration limits, the error between the trajectory profile and the reference line, the deviation from road boundaries, and the deviation from the lateral acceleration limits.

$$\begin{cases} f_v\left(x, v_{Lp}\right) = b_3\left(v_{Lp}\right)x^3 + b_2 v_{Lp}\left(v_{Lp}\right)x^2 + b_1 x + b_0 \\ f_{ax}\left(x, v_{Lp}\right) = \left(3b_3\left(v_{Lp}\right)x^2 + 2b_2 x + b_1\right) \cdot f_v\left(x, v_{Lp}\right) \\ f_y\left(x, y_{Lp}\right) = a_4\left(y_{Lp}\right)x^4 + a_3\left(y_{Lp}\right)x^3 + a_2 x^2 + a_1 x + a_0 \\ f_{ay}\left(x, v_{Lp}, y_{Lp}\right) = \dfrac{\left(f_v\left(x, v_{Lp}\right)\right)^2}{1 + \left(f_{dy}\left(x, y_{Lp}\right)\right)^2} \cdot \dfrac{d}{dx} f_{dy}\left(x, y_{Lp}\right) \end{cases} \tag{18}$$

$$g_v\left(x, v_{Lp}\right) = \begin{cases} 1 & \left(f_v\left(x, v_{Lp}\right) > v_{ref}\left(x\right)\right) \\ 0 & \left(0 < f_v\left(x, v_{Lp}\right) \le v_{ref}\left(x\right)\right) \end{cases} \tag{19}$$

$$g_{ax}\left(x, v_{Lp}\right) = \begin{cases} \left(f_{ax}\left(x, v_{Lp}\right) - A_{xmax}\right)^2 & \left(f_{ax}\left(x, v_{Lp}\right) > A_{xmax}\right) \\ 0 & \left(A_{xmin} \le f_{ax}\left(x, v_{Lp}\right) \le A_{xmax}\right) \\ \left(f_{ax}\left(x, v_{Lp}\right) - A_{xmin}\right)^2 & \left(f_{ax}\left(x, v_{Lp}\right) < A_{xmin}\right) \end{cases} \tag{20}$$

$$g_y\left(x, y_{Lp}\right) = \begin{cases} \left(f_y\left(x, y_{Lp}\right) - Y_{road_left}\left(x\right)\right)^2 & \left(f_y\left(x, y_{Lp}\right) > Y_{road_left}\left(x\right)\right) \\ 0 & \left(Y_{road_right}\left(x\right) \le f_y\left(x, y_{Lp}\right) \le Y_{road_left}\left(x\right)\right) \\ \left(f_y\left(x, y_{Lp}\right) - Y_{road_right}\left(x\right)\right)^2 & \left(f_y\left(x, y_{Lp}\right) < Y_{road_right}\left(x\right)\right) \end{cases} \tag{21}$$

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$$g_{ay}\left(x, v_{Lp}, y_{Lp}\right) = \begin{cases} 1 & \left(\left|f_{ay}\left(x, v_{Lp}, y_{Lp}\right)\right| > A_{ymax}\right) \\ 0 & \left(0 \le \left|f_{ay}\left(x, v_{Lp}, y_{Lp}\right)\right| \le A_{ymax}\right) \end{cases} \tag{22}$$

4. Collision avoidance simulations

4.1 Vehicle control model

The vehicle used in this study is an automated driving vehicle, and no driver operation is involved. The control targets in the controller are jerk and yaw angular acceleration. The target jerk curve $\dddot{v}(x)$ is determined from the second-order derivative of time t, and the control input is calculated from the target jerk at $x = 0$ in Eqs. (23) and (24).

$$\dddot{v}(x) = \frac{d^2}{dt^2} f_v(x) = f_{ddv}(x)\left(\frac{dx}{dt}\right)^2 \tag{23}$$

$$\dddot{v}_{input} = \dddot{v}(0) = 2b_2 v_{ego}^2 \tag{24}$$

Similarly, the target angular acceleration of the generated curve is calculated using Eq. (25), and the corresponding control input is computed from the target angular acceleration curve at $x = 0$ in Eq. (26).

$$\begin{aligned} \ddot{\psi}(x) &= \frac{d}{dt}\dot{\psi}(x) \\ &= \frac{d}{dt}\left\{\frac{dx}{dt}\cdot\frac{1}{1+\left(f_{dy}(x)\right)^2}\cdot f_{ddy}(x)\right\} \\ &= \frac{d^2x}{dt^2}\cdot\frac{f_{ddy}(x)}{1+\left(f_{dy}(x)\right)^2} + \left(\frac{dx}{dt}\right)^2 \frac{\left\{1+\left(f_{dy}(x)\right)^2\right\} f_{ddy}(x) - 2f_{dy}(x)\left(f_{ddy}(x)\right)^2}{\left\{1+\left(f_{dy}(x)\right)^2\right\}^2} \end{aligned} \tag{25}$$

$$\begin{aligned} \ddot{\psi}_{input} &= \frac{d}{dt}\dot{\psi}(0) \\ &= \frac{2a_2}{1+a_1^2}\dot{v}_{ego} + \frac{6\left(1+a_1^2\right)a_3 - 8a_1a_2^2}{\left(1+a_1^2\right)^2} v_{ego}^2 \end{aligned} \tag{26}$$

4.2 Simulations

4.2.1 Preparation

The simulations were conducted using MATLAB to demonstrate the effectiveness of the proposed risk prediction method and simultaneous optimization-based path generation method. An autonomous vehicle was simulated to pass through a narrow blind T-intersection, where the crossing had a width of 5.0 m. It was a two-way narrow road without center lines, and the speed limit was set to 30 km/h. The simulation adopted a 3-DoF point mass vehicle model. This model simplifies the vehicle motion into yaw dynamics and longitudinal and lateral directions. The vehicle was controlled to follow the target acceleration profile and trajectory curve introduced in the previous sections. The ego autonomous vehicle was equipped with an AEBS that imitates the functionality of a conventional AEBS. The system activates emergency brake when a moving object collides with the front face of the ego vehicle under the assumption that both objects maintain uniform linear motion, and when the TTC of the ego vehicle becomes shorter than a predefined threshold t_{tcEB}. When the emergency brake is applied, the vehicle decelerates with a constant jerk j_{EB_lim} and its acceleration is saturated at a_{EB}. Table 1 presents the configurations of the vehicle model, the autonomous driving system,

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the AEBS, and the darting-out object used in the simulation.

4.2.2 Vehicle navigation and control simulations

A vehicle automatically drove through the course without any driver input. In this simulation, three road widths were prepared: 4.0 m, 6.0 m, and 8.0 m. Risk-predictive autonomous driving simulations were conducted under each of these conditions in MATLAB. The simulation calculations were performed in a real-time environment with average update frequencies 130.3 Hz (Intel Core i9-10900K CPU @ 3.70GHz, RAM 32GB, NVIDIA GeForce RTX 3070).

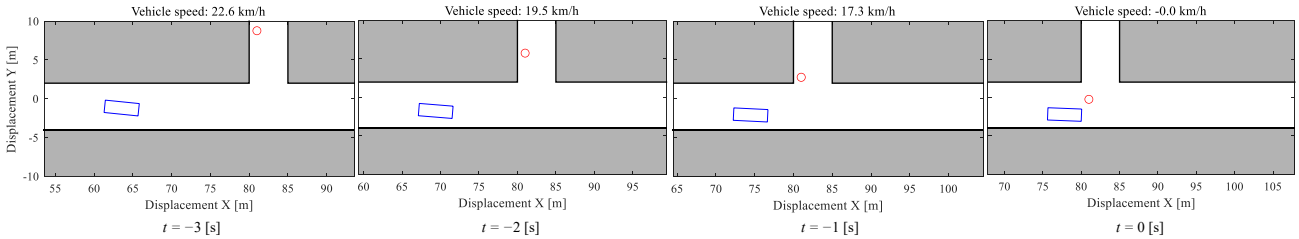
Figure 6 (a) and (b) show the timelapse snapshots of the bird-eye-view of the simulations. The results indicated that our proposed method controlled the vehicle to take an appropriate lateral gap and decelerate in advance, allowing the vehicle to effectively avoid the collision with the darting-out object. On the other hand, the vehicle without the proposed method could not avoid the collision even with AEBS engagement.

Figure 7 (a) to (d) show the simulations results under three road-width conditions. As shown in the figures, the proposed system successfully generated safe speed and trajectory profiles simultaneously, while avoiding high-risk areas with moderate and smooth acceleration and steering maneuvers across all the road widths. Under the narrow roads conditions (4.0 m and 6.0 m), the road did not provide sufficient space to avoid entering the predicted dart-out area (green hatched region). As a result, the vehicle decelerated to calculated safe speed. In these cases, the vehicle successfully avoided frontal collisions with the assumed darting-out object through AEBS intervention, as shown in Fig. 7 (d). However, side-face collisions may not be avoided if the darting-out object does not perform any avoidance maneuver. Conversely, under the wider road conditions (8.0 m) in Fig. 7 (c), sufficient space was available to maintain distance from the blind spot, and the vehicle passed through the intersection without entering the side-collision-risk area.

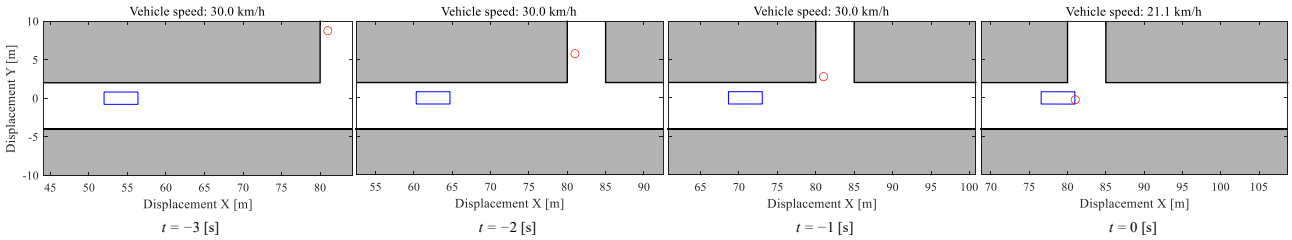
Table 1 Simulation parameters.

Parameters		Sign	Value	Unit
Ego vehicle	Length	L	4.40	m
	Width	W	1.60	m
Risk predictive safe speed and lateral gap calculation	Speed of virtual object	v_{vir}	3.00	m/s
	Predefined critical TTC	t_{crit}	1.50	s
	Minimum longitudinal acceleration	A_{x_min}	−2.94	m/s^2
	Maximum longitudinal acceleration	A_{x_max}	1.96	m/s^2
	Maximum absolute lateral acceleration	A_{y_max}	1.96	m/s^2
	Assumed AEBS latency	τ_{EB}	0.40	s
	Assumed AEBS acceleration	a_{EB_lim}	−7.85	m/s^2
	Darting-out lateral offset	D_{offset}	0.5	m
	Look-ahead time of path planning	t_p	3.0	s
	Weighting factors	w_1	500.0	-
		w_2	7000.0	-
		w_3	5000.0	-
		w_4	500.0	-
		w_5	10000.0	-
		w_6	100.0	-
AEBS	Minimum braking jerk by AEB	j_{EB_lim}	−15.0	m/s^3
	Minimum acceleration by AEB	a_{EB}	−9.00	m/s^2
	Sensor detection latency	τ_{sens}	0.30	s
	Threshold TTC for AEBS activation	t_{tcEB}	1.00	s
Darting out object	Darting-out speed	v_{obj}	3.0	m/s
	Darting-out position	X_{obj}	81.0	m

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(a) Collision avoidance by the proposed risk-prediction and navigation method.



(b) Without risk prediction driving and collision-avoidance attempt by AEBS.

Fig. 6 The timelapse snapshots of the simulations: (a) Collision avoidance by the proposed risk-prediction and navigation method, and (b) Driving without risk prediction (maintaining the speed and lateral position) and collision-avoidance attempt by AEBS. The time t was calibrated to the moment when the darting-out object reaches the collision point in Fig. 6 (b).

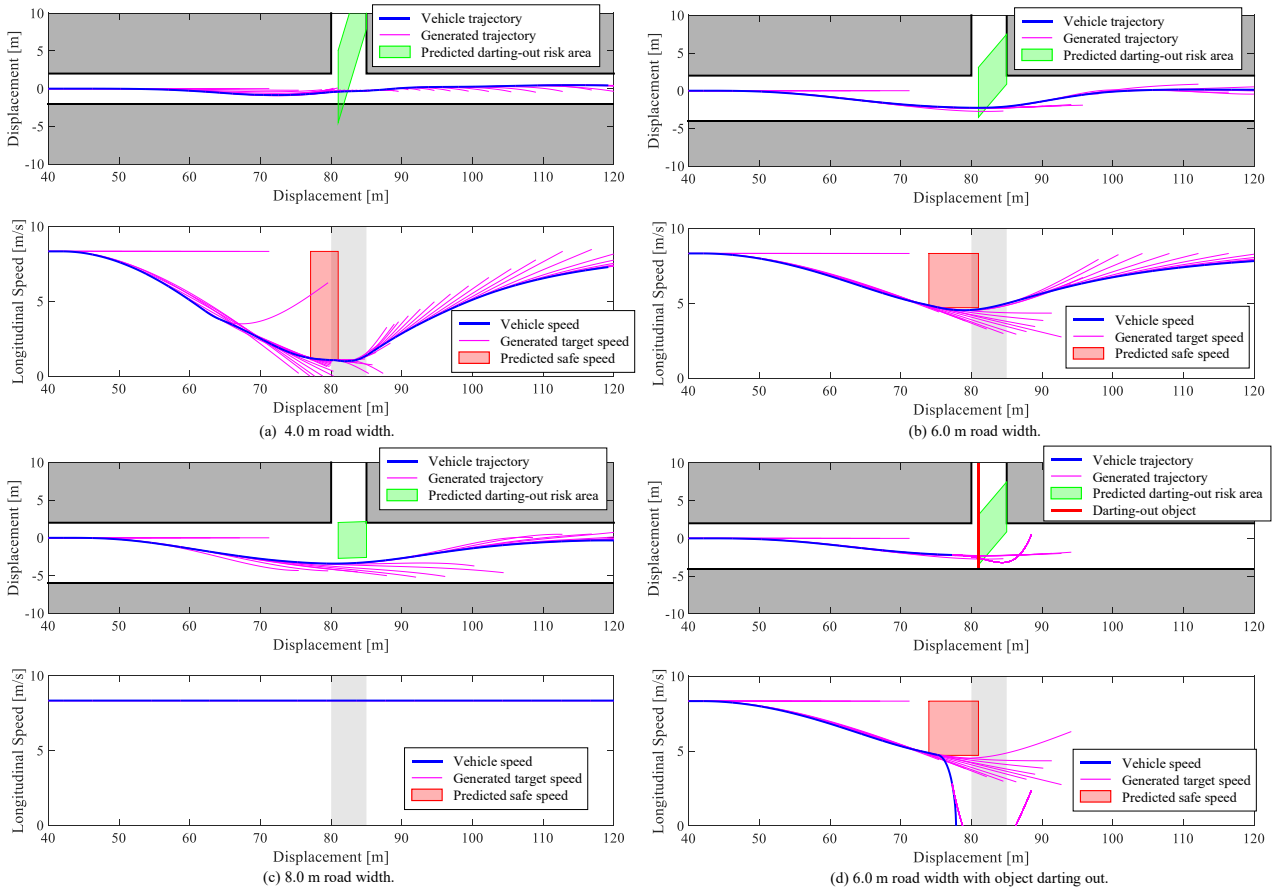


Fig. 7 Generated curves and simulated driving paths of the ego vehicle under each road width condition. (a) 4.0 m road width, (b) 6.0 m road width, (c) 8.0 m road width, and (d) 6.0 m road width when an object crossed the road in front of the vehicle.

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4.2.3 Collision avoidance performance evaluation for darting-out pedestrian

To evaluate the collision-avoidance performance against darting-out pedestrians, we conducted comprehensive simulations under various darting-out conditions. Road widths ranging from 4.0 to 9.0 m were prepared, and a pedestrian was set to cross the intersection at predetermined timings. The darting-out timing, referred to as the "time offset," was varied from −3.0 to 11.0 seconds, calibrated such that 0.0 seconds corresponds to the moment when the darting-out object would collide with the front-left corner of the ego vehicle if neither AEB nor safe speed/lateral gap control were applied. As the time offset increases, the object darts out at a later timing.

Figure 8 shows the collision-avoidance results of the comprehensive simulations. The results revealed that no cases involving front-face collisions of the ego vehicle collided were observed, since all such cases were avoided by AEB engagement. Next, 200 collisions occurred out of 5781 simulation cases. In these cases, the darting-out object collided with the side-face of the ego vehicle, which cannot be prevented solely through AEB activation. However, 197 of the 200 collisions occurred with a TTC of 1.5 s or more, measured at the moment the darting-out became visible to the ego vehicle's sensor position (TTC-visible). Only 3 cases had a TTC-visible shorter than 1.5 s, with the minimum TTC-visible being 1.4 s. From the simulation parameters, avoiding entry into the collision-risk area required the ego vehicle to maintain a lateral distance greater than 7.4 m, and the simulation results also indicated that no collision occurred at road width exceeding 7.4 m. From those results, the proposed risk-prediction and control method successfully navigated the vehicle to safe states and effectively avoided collisions across a wide range of road width and darting-out timing.

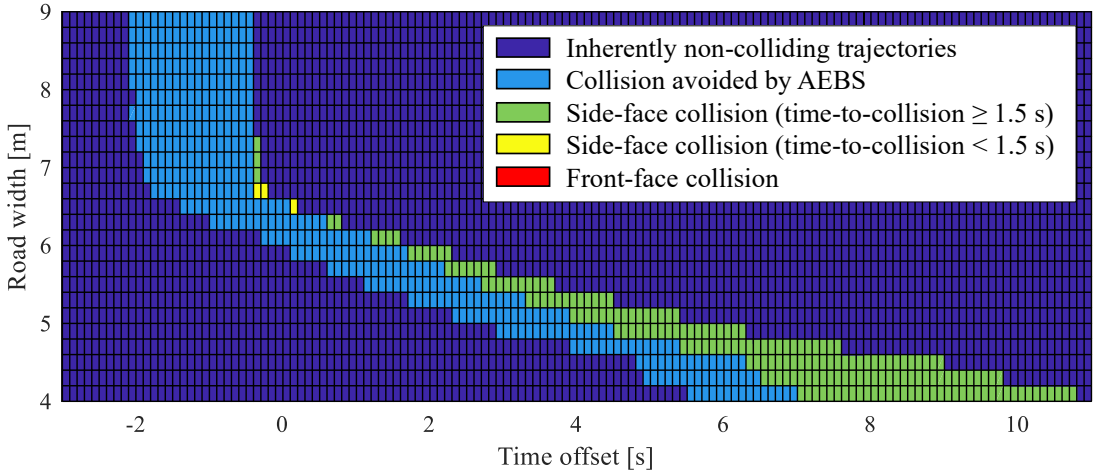


Fig. 8 Results of comprehensive simulations. The blue cells indicate cases with no possibility of collision, the light blue cells indicate cases avoided by AEB engagement, the green cells indicate side-face collision where TTC-visible ≥ 1.5 s at object visibility, the yellow cells indicate side-face collision cases with TTC-visible < 1.5 s, and the red cells indicate cases where the front-face of the ego vehicle collided with the object.

4.2.4 Effect of prediction accuracy on darting-out speed parameter

This subsection examines how the prediction accuracy of the virtual dart-out speed v_{vir} affects the collision-avoidance performance of the proposed method. By fixing the road width as 7.0 m, we conducted a comprehensive simulation in which the predicted dart-out speed of the virtual object and the actual dart-out speed were independently varied from 1.0 to 10.0 m/s, and the dart-out timing was set from −3.0 to 11.0 seconds.

Figure 9 shows the simulation results. When the predicted dart-out speed v_{vir} was higher than the actual speed v_{obj}, the number of collision cases increased, but all TTC-visible values were greater than 1.5 s. This is because a higher predicted dart-out speed yields a lower calculated safe speed, resulting in slower vehicle traversal through the intersection. Although this increased the number of collision cases, it also provides a longer TTC for the darting-out object. On the other hand, when the predicted dart-out speed v_{vir} was lower than the actual speed v_{obj}, some collisions occurred with TTC-visible values shorter than 1.5 s. These results indicated that the accuracy of the predicted dart-out speed affects collision-avoidance performance; however, overestimating the dart-out speed does not significantly degrade system safety. Therefore, the virtual dart-out speed parameter allows the system to be configured with different levels of conservativeness under predefined safety constraints. Ideally, the virtual dart-out speed should be set to accurately reflect

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actual darting-out behavior; however, achieving such accuracy would require sufficient statistical observations of darting-out characteristics in occluded environments, as well as a reliable theoretical model to describe this behavior, both of which remain challenging under current knowledge.

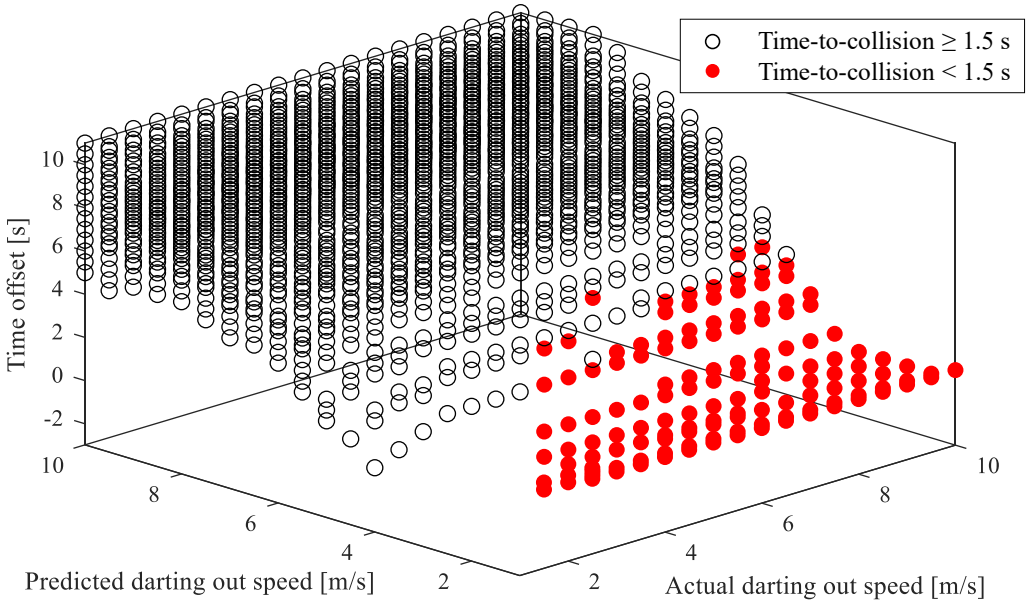


Fig. 9 Results of simulations when the predicted dart-out speed v_{vir} deviated from the actual dart-out speed v_{obj}. The white circles indicate collided cases with TTC-visible ≥ 1.5 s; the red-filled circles indicate TTC-visible < 1.5 s.

5. Conclusions

This paper focused on the straight-road driving on urban narrow roads and proposed risk prediction and continuous path generation methods that achieve safe vehicle control for autonomous driving and advanced driver assistance systems. The proposed framework enables autonomous vehicles to safely navigate scenarios involving blind spots, balancing safety and driving comfort.

The main contributions of this research are summarized as follows:

- Theoretical definitions of safe speed and lateral distance that guarantee collision avoidance through the emergency avoidance system.
- Comprehensive simulations demonstrating that the proposed method effectively avoided darting-out events.
- Polynomial-based curve generation method enabling the autonomous vehicle to navigate safely without entering collision-risk area.

Simulation results confirmed that the proposed framework can simultaneously determine safe speed and lateral distance in real time, ensuring both safety and mobility in narrow urban environments.

Compared with existing occlusion-aware approaches such as potential field-based methods and Responsibility-Sensitive Safety, the proposed method formulates safety using a geometric risk-prediction framework with closed-form computation. This enables efficient real-time evaluation of safety constraints while maintaining deterministic safety assurance in narrow urban environments. The proposed approach is particularly suitable for practical autonomous driving and advanced driver assistance systems and can serve as a complement to existing safety functionalities in autonomous driving applications. Furthermore, the proposed algorithm can be integrated into existing AD/ADAS architectures using onboard sensors and vehicle control units without requiring additional hardware.

Future work will extend the proposed method to more complex road geometries, including curved roads, multiple blind spots, and dynamic traffic participants, and validate its performance through Vehicle-in-the-Loop experiments.

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